

MATSUSHIMA AB, Japan

WMO: 475910

Lat: 38.403N

Lon: 141.212E

Elev: 17

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.6	24.5	13.7	11.0	29.0	15.5	12.1	30.2	30.6	36.2	28.2	36.5	6.0	300	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.0	85.8	76.0	82.7	75.6	80.7	74.8	78.6	82.7	77.4	81.3	76.2	79.8	10.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.2	141.4	80.9	75.6	134.1	79.5	75.0	131.5	79.0	42.1	83.2	40.6	80.9	39.4	79.8	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.1	23.1	20.1	DB	15.9	91.6	3.3	3.0	13.6	93.8	11.6	95.5	9.7	97.2	7.3	99.4
			WB	14.7	79.8	3.0	1.7	12.6	81.1	10.8	82.0	9.1	83.0	7.0	84.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	53.6	33.5	34.6	39.8	49.1	58.6	65.3	72.6	75.3	69.3	58.8	47.4	37.9	(6)
(7)		DBStd	15.34	4.38	4.58	5.03	5.65	4.98	4.30	4.90	4.35	5.32	5.41	5.76	4.94	(7)
(8)		HDD50	1838	511	432	317	83	2	0	0	0	0	3	115	375	(8)
(9)		HDD65	4927	976	852	780	476	204	47	4	0	19	203	527	839	(9)
(10)		CDD50	3166	0	0	2	57	270	460	699	783	580	276	38	1	(10)
(11)		CDD65	780	0	0	0	0	7	57	238	319	148	11	0	0	(11)
(12)		CDH74	3728	0	0	0	2	28	108	1076	1892	604	18	0	0	(12)
(13)		CDH80	659	0	0	0	0	3	11	208	368	68	1	0	0	(13)
(14)	Wind	WSAvg	8.4	9.8	10.2	10.2	9.7	8.5	7.4	6.7	6.9	7.1	7.5	8.1	9.2	(14)
(15)	Precipitation	PrecAvg	45.4	1.6	1.5	2.7	3.5	4.0	5.0	6.1	4.9	6.3	5.0	2.9	1.9	(15)
(16)		PrecMax	60.2	5.4	3.5	6.4	6.5	8.0	10.5	12.9	18.0	18.7	14.8	6.1	7.6	(16)
(17)		PrecMin	34.8	0.3	0.2	0.2	0.3	1.4	2.5	2.1	0.8	0.5	0.9	0.5	0.6	(17)
(18)		PrecStd	6.7	1.3	0.8	1.5	1.8	1.7	1.8	3.0	4.0	3.9	3.4	1.7	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.1	53.5	60.7	70.1	77.9	81.0	89.5	91.0	84.6	76.6	66.0	57.2	(19)
(20)			MCWB	43.9	44.8	48.1	54.5	63.1	68.3	76.5	76.5	74.2	65.7	56.6	51.7	(20)
(21)		2%	DB	46.5	48.6	55.4	64.8	72.0	77.0	84.5	86.2	82.0	73.0	62.7	53.2	(21)
(22)			MCWB	40.5	42.1	46.8	53.6	59.9	68.4	75.7	76.9	74.7	65.5	56.7	47.3	(22)
(23)		5%	DB	44.2	46.2	52.0	61.1	69.4	73.8	82.2	84.0	79.1	70.0	60.6	49.9	(23)
(24)			MCWB	38.5	40.2	44.9	52.3	59.4	66.7	75.1	76.5	73.6	63.6	54.8	44.2	(24)
(25)		10%	DB	41.3	43.1	49.8	58.8	66.3	71.7	79.1	82.0	77.2	68.1	58.6	46.7	(25)
(26)			MCWB	36.0	37.4	43.7	51.3	59.1	66.2	74.2	76.3	72.5	61.9	52.9	41.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.5	47.3	51.8	59.7	65.7	72.7	79.4	80.1	78.6	70.8	60.2	53.1	(27)
(28)			MCD B	47.5	50.7	57.3	64.3	71.7	77.6	84.7	84.5	82.0	73.3	63.0	56.1	(28)
(29)		2%	WB	41.0	43.4	49.0	56.6	63.6	70.4	77.6	79.1	76.9	67.4	57.9	48.0	(29)
(30)			MCD B	44.9	47.7	52.7	61.7	68.5	74.5	82.6	83.1	79.8	70.5	61.4	51.6	(30)
(31)		5%	WB	38.7	40.5	46.6	54.7	62.0	68.9	76.2	78.1	75.6	65.2	55.6	44.5	(31)
(32)			MCD B	42.8	45.0	50.8	59.4	66.3	72.4	80.6	81.9	78.5	69.2	59.5	48.8	(32)
(33)		10%	WB	36.6	37.9	44.2	52.7	60.3	67.3	74.8	77.0	73.7	63.0	53.0	41.7	(33)
(34)			MCD B	41.0	42.7	48.6	57.2	64.3	70.6	78.3	80.6	76.7	67.2	57.3	45.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.7	14.8	16.3	16.3	13.3	10.4	9.5	10.0	11.9	15.2	16.0	13.9	(35)
(36)			MCD BR	16.8	18.4	19.8	20.4	18.2	14.0	14.0	13.6	13.0	15.9	18.7	17.4	(36)
(37)			MCWBR	12.4	13.4	13.9	12.2	9.7	7.9	7.2	7.1	7.7	10.4	13.8	13.0	(37)
(38)		5% WB	MCD BR	16.2	17.6	18.1	16.8	13.3	11.3	11.6	10.7	10.0	13.0	17.3	17.5	(38)
(39)			MCWBR	12.8	14.0	14.6	11.5	8.6	7.4	6.7	6.5	7.1	10.2	14.2	13.9	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	0.369	0.414	0.488	0.537	0.558	0.552	0.561	0.547	0.480	0.440	0.401	0.377	(41)	
(42)		taud	2.171	2.045	1.896	1.839	1.851	1.944	1.983	2.011	2.163	2.175	2.188	2.183	(42)	
(43)		Ebn at Noon	246	250	242	239	237	238	235	234	243	239	233	233	(43)	
		Edh at Noon	35	45	57	64	65	59	56	54	43	39	34	32	(43)	
(44)	All-Sky Solar Radiation	RadAvg	658	898	1190	1481	1599	1544	1405	1392	1155	943	698	561	(44)	
(45)		RadStd	38	62	74	142	156	95	213	179	106	95	44	43	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.67	N/A	N/A	N/A	N/A

Nomenclature: See separate page